Lab 4.1 - WBMUD Enterprise Network & Security Lab

Title Page

Name: Nada Alaiban

Course: IT 520

Lab Title: Lab 4.1 - WBMUD Enterprise Network & Security Lab

Date: 4-14-2026

Introduction

This lab focused on designing and securing a segmented enterprise network that separates IT and OT networks. The lab demonstrated OSI layers, routing between networks, and secure protocol implementations such as HTTPS and SSH. The goal of the lab was to understand network architecture, protocol security, and secure remote management in enterprise environments.

Part 1 – Network Architecture & OSI Layers 1–3

Topology Description

The WBMUD network was designed using three main operational segments and one DMZ segment. The IT network represented the office environment, the OT network represented industrial control and monitoring systems, and the Remote Pump Station represented a remote operational site connected through WAN links. A DMZ attacker node was also included for security testing. Each network was assigned its required IP range, and routers were connected using serial links to support communication between segments.

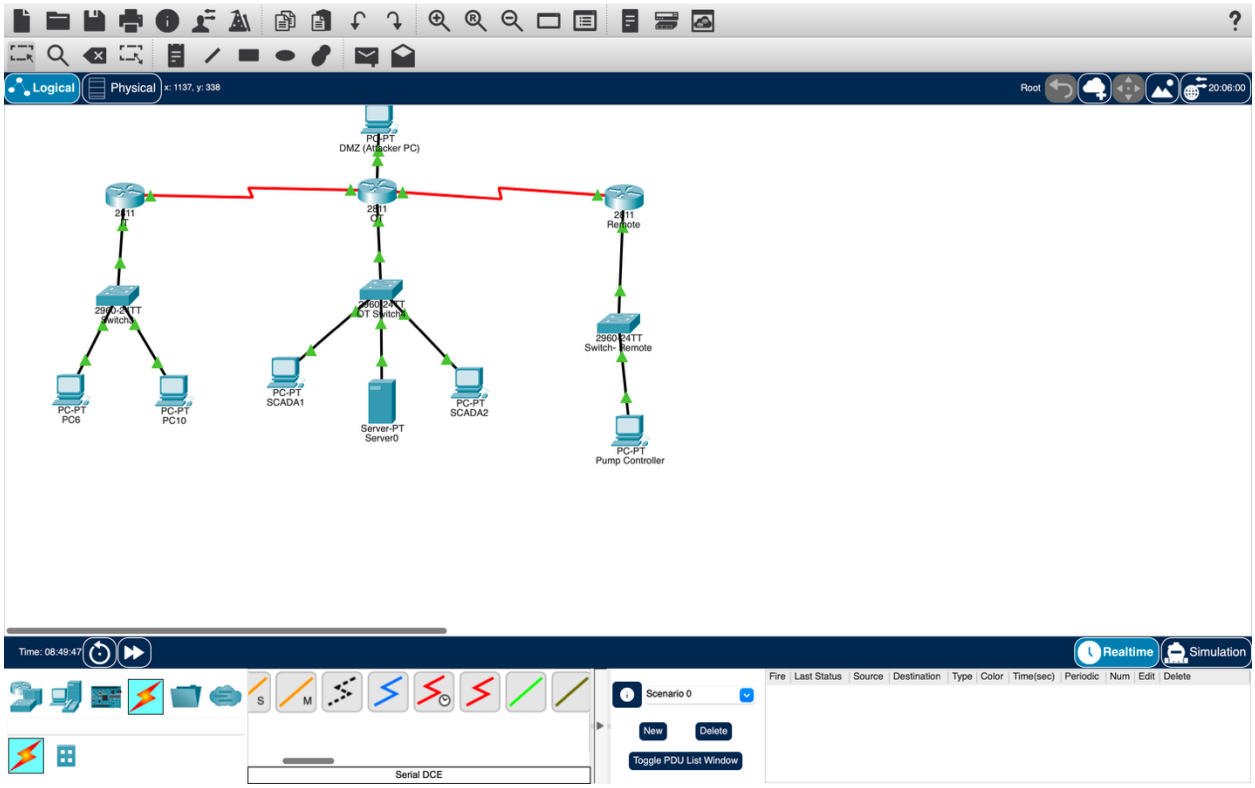

Figure 1. Full topology view.

OSI Layer 1 Reflection

The Physical layer is responsible for transmitting raw bits over a physical medium. In this lab, cables, ports, and powered devices represent Layer 1 because they provide the physical connection required for communication. This layer does not handle IP or MAC addressing. It focuses only on the transmission of bits between devices.

OSI Layer 2 Reflection

OSI Layer 2 is the Data Link layer, responsible for MAC addresses and switching. Switches forward frames based on MAC addresses. The MAC address table shows how switches learn device addresses and forward traffic within the local network.

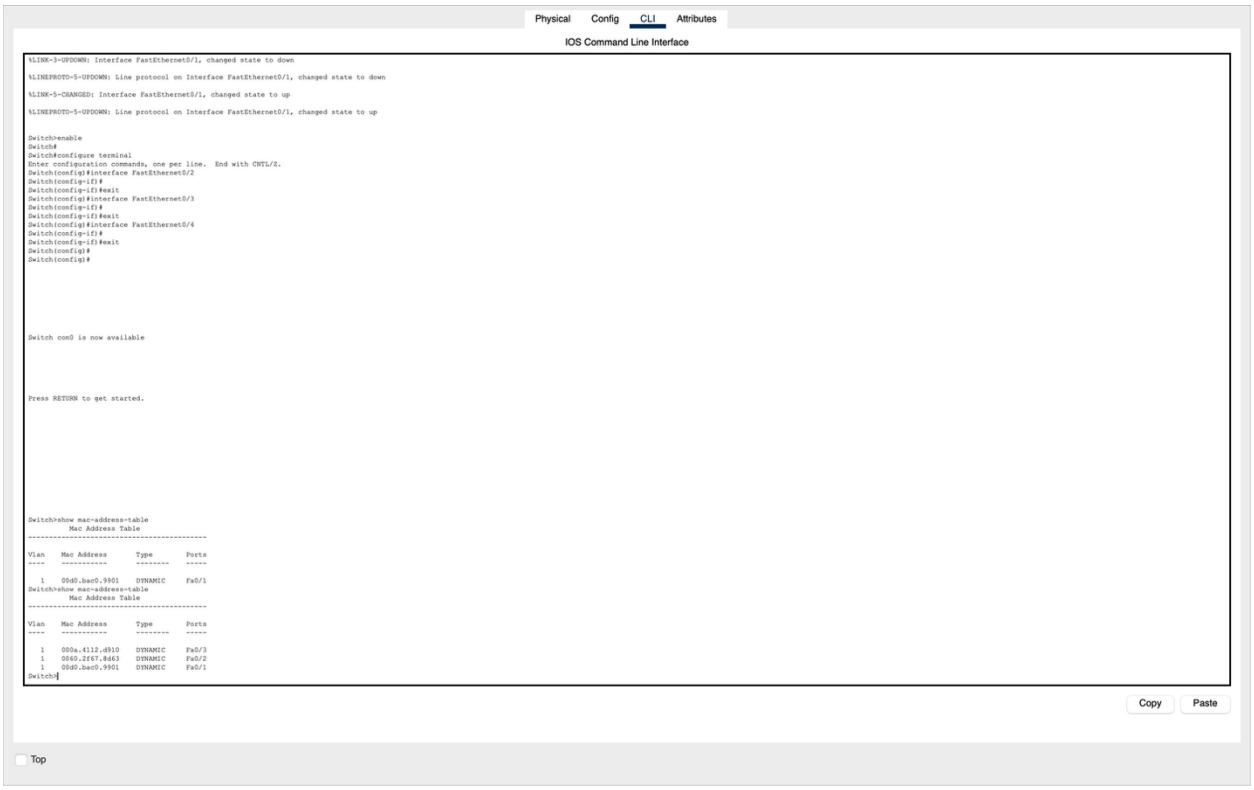

Figure 2. Switch MAC address table showing dynamically learned MAC addresses and associated switch ports.

OSI Layer 3 Reflection

OSI Layer 3 is the Network layer, responsible for IP addressing and routing. Routers use routing tables to forward packets between different networks. In this lab, routers allowed communication between IT, OT, and Remote networks using IP routing.

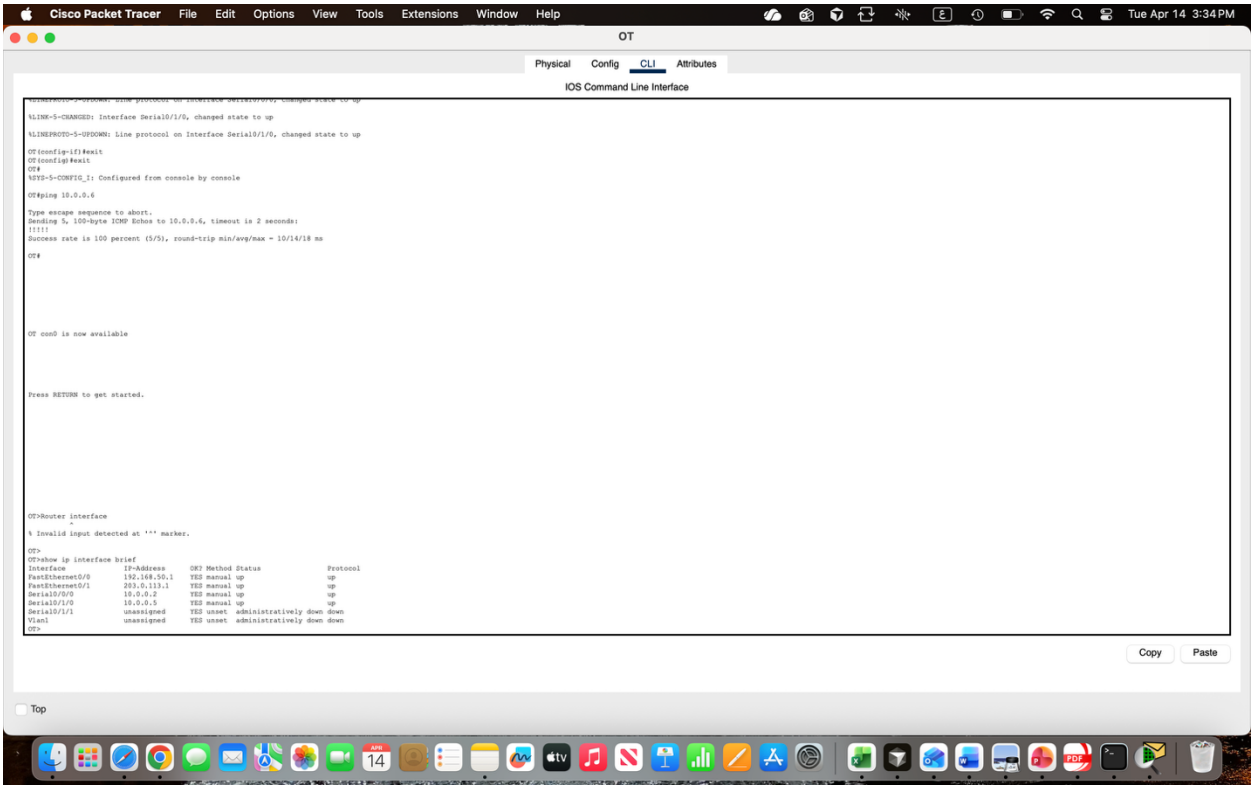

Figure 3. Router interface summary showing assigned IP addresses and operational interface status.

OSI Layer 3 Reflection

The Network layer is responsible for logical addressing and routing between different networks. In this lab, IP addressing, subnet masks, default gateways, and static routes were all configured at Layer 3. Routers used this information to forward packets between the IT, OT, and Remote Pump Station networks. Without Layer 3 routing, devices would only be able to communicate within their own subnet.

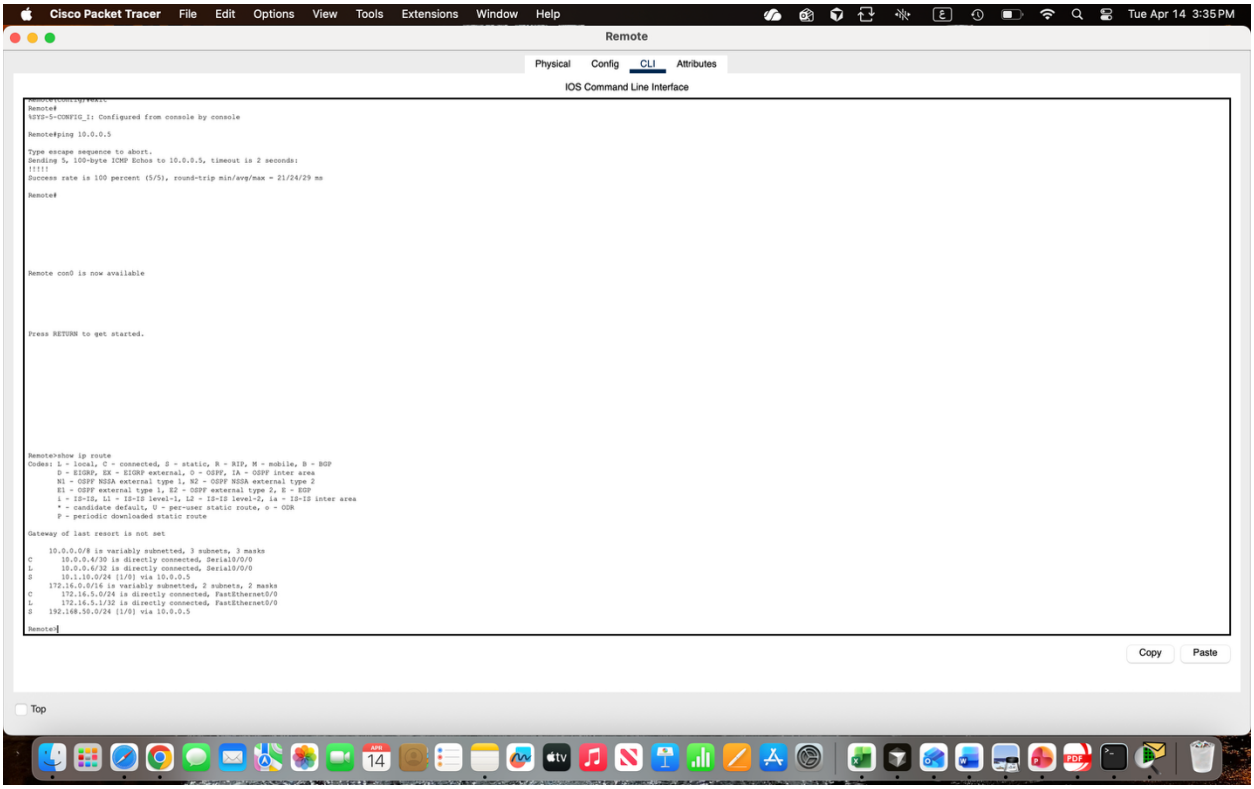

Figure 4. Routing table showing directly connected networks and configured static routes between WBMUD segments.

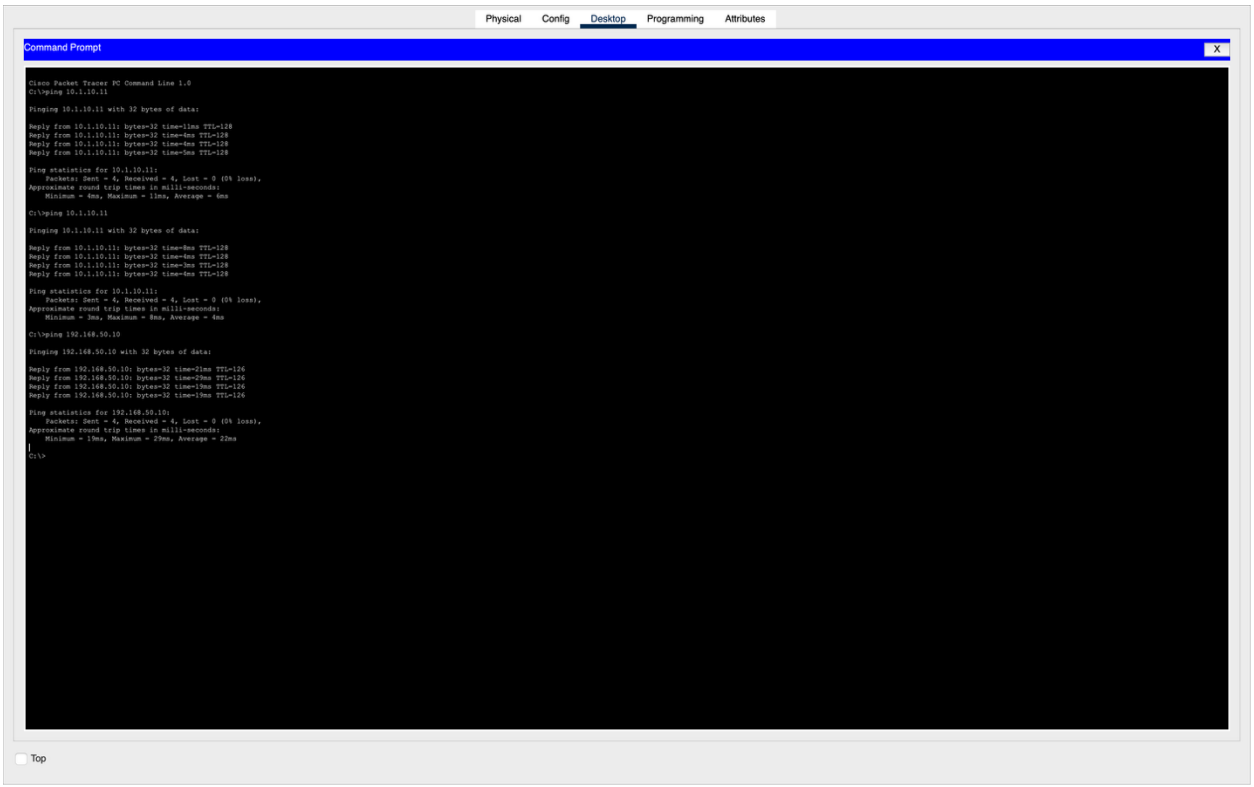

Figure 5. Connectivity verification from the IT workstation showing successful pings to a local IT host, the OT web server, and the Remote Pump Station controller.

Troubleshooting Notes

Diagnosis:

Initial connectivity problems were caused by incorrect IP addressing and incomplete routing verification. In the earlier version of the topology, the OT web server was configured with an incorrect IP address, and some required routing evidence was not included in the report.

Resolution:

The OT web server IP address was corrected to 192.168.50.10, and router interface and routing table verification were completed using show ip interface brief and show ip route. After correcting the addressing and confirming static routes, all required ping tests from the IT workstation were successful.

Part 2 – Protocol Security & OSI Layers 4–7

This section analyzes the security implications of TCP vs UDP, HTTP vs HTTPS, and SSH vs Telnet in the WBMUD network.

TCP vs UDP Security Explanation

TCP provides greater security visibility because it uses a three-way handshake: SYN, SYN-ACK, and ACK. This makes TCP communication stateful, which allows firewalls and intrusion detection systems to track sessions more effectively. UDP, by contrast, is connectionless and sends datagrams without establishing a session first. As a result, UDP offers less visibility and less reliable state tracking for security monitoring.

OSI Layer 4 Reflection

TCP and UDP operate at OSI Layer 4, the Transport layer. TCP provides reliable communication using acknowledgments, sequencing, and connection establishment, but it is slower. UDP is faster because it does not establish a connection or guarantee delivery, but it is less reliable. From a security perspective, TCP is easier for firewalls and IDS/IPS systems to track because it is stateful, while UDP is stateless and harder to monitor.

HTTP vs HTTPS Security

HTTPS is more secure than HTTP because it uses TLS/SSL encryption to protect data in transit between the client and the server. HTTPS provides encryption, integrity protection, and server authentication using digital certificates, which helps prevent man-in-the-middle attacks. HTTP sends data in clear text, making it easy for attackers to intercept or modify the data.

Figure 6. Browser access to the OT web server using both HTTP and HTTPS protocols.

OSI Layer Mapping for Encryption

TLS/SSL encryption used in HTTPS is generally considered part of the Application layer (Layer 7) of the OSI model, sometimes described across Layers 5–7. TLS operates on top of TCP and encrypts application data before it is transmitted over the network. This protects the data while it is transported by TCP at the Transport layer.

SSH vs Telnet Security

SSH replaced Telnet because Telnet sends usernames, passwords, and commands in clear text, which can be intercepted by attackers. SSH encrypts the entire remote management session, including login credentials and commands, providing confidentiality and integrity. This makes SSH much more secure and protects against eavesdropping and session hijacking. I used a serial DCE cable.

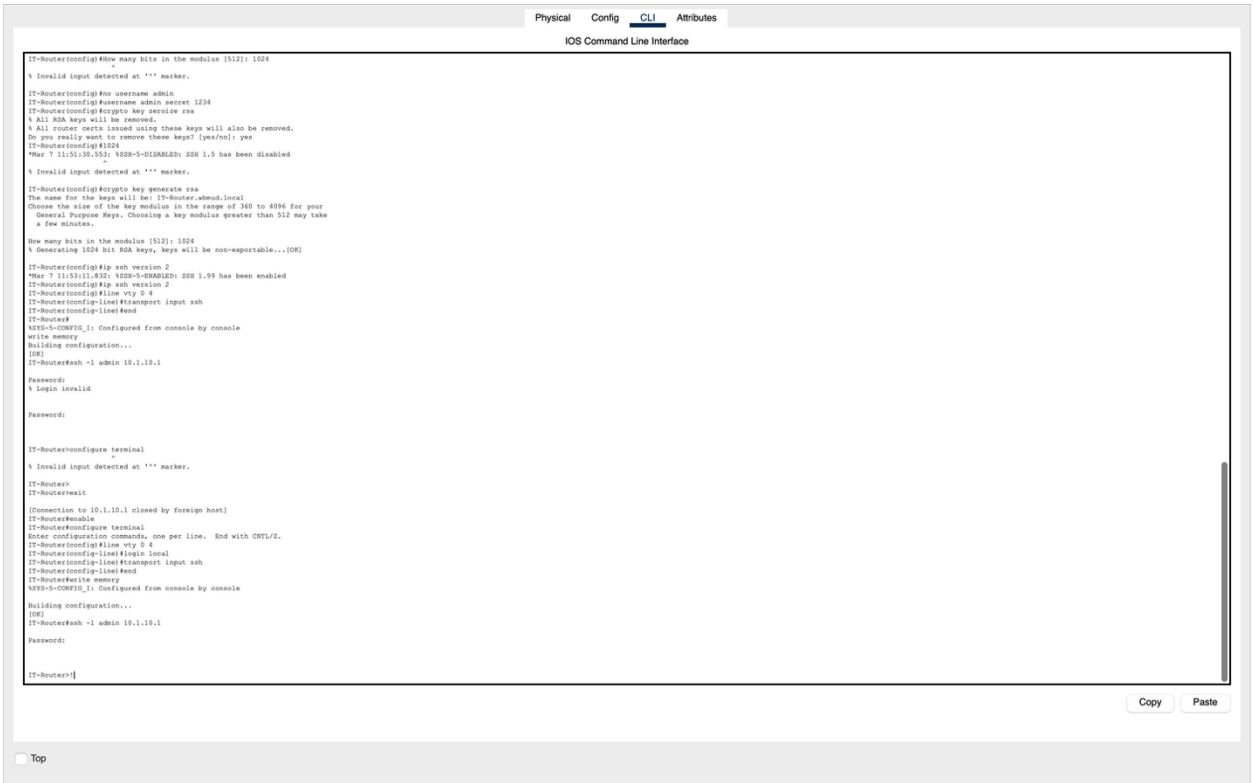

Figure 7. Successful SSH session from the IT workstation to the IT router, demonstrating encrypted remote management access.

DMZ Attacker Node

The DMZ Attacker PC was added to simulate an external threat to the network. This device represents a potential attacker located outside the internal IT and OT networks. It was configured with the IP address **203.0.113.100** and connected directly to the router to create a separate DMZ segment.

The purpose of this setup is to test network security and evaluate how the system responds to external access attempts. By isolating the attacker in a DMZ, the network design ensures that internal systems remain protected while still allowing controlled testing.

A successful ping test between the attacker PC and the router confirmed proper connectivity and correct configuration of the DMZ network.

Conclusion

This lab demonstrated how enterprise networks use segmentation and secure protocols to protect critical infrastructure systems. The lab showed the importance of using secure protocols such as HTTPS and SSH instead of HTTP and Telnet. The key takeaway is that protocol security and OSI layer understanding are essential for designing secure enterprise IT and OT networks

GitHub Evidence Locker
The supporting evidence for this lab, including screenshots and documentation, is stored in the following GitHub repository:
https://github.com/N-alaiban/WBMUD-Evidence-Locker